# STRATEGIC APPLICATION OF ARTIFICIAL INTELLIGENCE IN CAD

*A Project Report Submitted to*
*Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal*
*towards the partial fulfillment of the degree of Bachelor of Technology*
*in Mechanical Engineering*

**2026 - 2027**


**Supervised by:**

Dr. B. R. Rawal

Professor and Head

Department of Mechanical Engineering

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**SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE (M.P.)**

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A Govt. Aided Autonomous Institute, Affiliated to RGPV, Bhopal

**DEPARTMENT OF MECHANICAL ENGINEERING**

**2026 - 2027**

# RECOMMENDATION

**We are pleased to recommend that the project work entitled *Strategic Application of Artificial Intelligence in CAD* submitted by Ankur Dodke, Ansh Yadav, Atishay Jain, Bhavya Shrivastava, Dharmendra Gupta may be accepted in partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the session 2026 - 2027.**

**Dr. B. R. Rawal**
**Professor and Head**
**Department of Mechanical Engineering**

**Head of Department**
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**SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE (M.P.)**

A Govt. Aided Autonomous Institute, Affiliated to RGPV, Bhopal

**DEPARTMENT OF MECHANICAL ENGINEERING**

**2026 - 2027**

# CERTIFICATE

**This is to certify that the Project entitled *Strategic Application of Artificial Intelligence in CAD* submitted by Ankur Dodke, Ansh Yadav, Atishay Jain, Bhavya Shrivastava, Dharmendra Gupta is accepted in partial fulfillment of the degree of Bachelor of Technology in Mechanical Engineering of Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal during the session 2026 - 2027.**

| | |
|---|---|
| **Internal Examiner** | **External Examiner** |
| **Date:** ..... | **Date:** ..... |

# DECLARATION

We, the undersigned students of the final year Bachelor of Technology course in Mechanical Engineering, Department of Mechanical Engineering, SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE (M.P.), declare that the project entitled *Strategic Application of Artificial Intelligence in CAD* is our own work conducted under the supervision of Dr. B. R. Rawal, Professor and Head, Department of Mechanical Engineering, Shri G. S. Institute of Technology and Science, Indore.

We affirm that this work is entirely original and has not been copied from any source. All sources consulted during the work have been duly acknowledged and cited in the references. Every design formula and empirical proportion used in the software developed for this project is attributed in the text to the published source from which it is taken, and every instance in which the software departs from a cited source is stated explicitly in Chapter 7 together with the reason for the departure.

We further declare that, to the best of our knowledge, this project work does not contain any part of any work which has been submitted for the award of any degree either in this University or in any other University or from any website without proper citation.

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|---|---|---|---|---|
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We thank the faculty of the Department of Mechanical Engineering for their observations during the review presentations, which repeatedly sharpened the scope of the work, and the Institute for providing the facilities and the academic environment in which the work was carried out.

We acknowledge the National Programme on Technology Enhanced Learning and the Indian Institute of Technology Kharagpur, whose openly published course material on the design of machine elements is the primary reference for this work, and the Internet Archive, through which the out of copyright texts by Unwin and by Leutwiler were obtained.

Finally, we thank our families and our classmates for their support during the course of this project.

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# ABSTRACT

Designing a standard machine joint is a repetitive calculation. For a knuckle joint the designer computes the rod diameter from direct tension, sizes the pin separately for double shear and for bending, adopts the larger of the two requirements, derives the remaining proportions from the rod diameter, and then verifies nine distinct failure modes against allowable stresses. A change in the applied load invalidates every one of those results and requires the entire sequence again. The drafting that follows is mechanical but unavoidable. Computer aided design software removes the labour of drawing but not the labour of calculation: a modeller regenerates whatever dimensions it is given and will not object if a pin is too slender to carry the bending moment placed upon it.

This project presents a software system that performs the complete sequence automatically from a requirement written in ordinary English. The user enters a sentence such as *knuckle joint for 50 kN*. A language model extracts the parameters from that sentence. All subsequent work is deterministic: the joint is sized from standard design formulas, each computed diameter is rounded up to the next preferred size, every applicable failure mode is verified against the allowable stress, any dimension that fails is increased until it passes, every component is generated as an exact solid model, and a first angle A3 drawing sheet is produced with dimension lines and a title block.

Six joints are implemented: the knuckle joint, the socket and spigot cotter joint, the sleeve and cotter joint, the gib and cotter joint, the hex bolted joint and the protected type rigid flange coupling. Together they apply 44 distinct failure checks. Every proportion and every check was compared against published design literature, and the comparison identified three errors in the sources themselves, including a worked example whose own answer exceeds its own allowable stress by 23 percent. The implementation is covered by 290 automated tests and is deployed as a web application requiring no installation.

The central design decision is that the language model reads the requirement only. It selects no dimension and performs no check. Interpreting a phrase is a problem of language; selecting a pin diameter is a problem of engineering with a single correct answer, and it remains with deterministic code whose every step can be traced to a printed page.

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# LIST OF SYMBOLS

| Symbol | Meaning | Unit |
|---|---|---|
| P | Applied axial load | N |
| T | Transmitted torque | N mm |
| D | Rod diameter | mm |
| d | Pin or spigot diameter | mm |
| $d_2$ | Outer diameter of the eye or socket | mm |
| t | Thickness of the eye, or of the cotter | mm |
| $t_1$ | Thickness of each fork plate | mm |
| b | Breadth of the cotter and gib | mm |
| $\sigma_t$ | Allowable tensile stress | MPa |
| σ_b | Allowable bending stress | MPa |
| σ_c | Allowable crushing stress | MPa |
| τ | Allowable shear stress | MPa |

# LIST OF ABBREVIATIONS

| Abbreviation | Expansion |
|---|---|
| CAD | Computer Aided Design |
| STEP | Standard for the Exchange of Product data |
| DXF | Drawing Exchange Format |
| FEA | Finite Element Analysis |
| NPTEL | National Programme on Technology Enhanced Learning |
| OCCT | Open CASCADE Technology |
| BIS | Bureau of Indian Standards |
| MPa | Megapascal, newton per square millimetre |

# CHAPTER 1

# INTRODUCTION

## 1.1 Background

A machine joint transmits load between two members. The standard joints taught in machine design, among them the knuckle joint, the cotter joint and the shaft coupling, are not designed by judgement. Each is sized by a fixed sequence of formulas and then verified against a fixed list of failure modes. The sequence is well established and appears in substantially the same form in every standard text on the subject.

Because the sequence is fixed, it is repetitive. A change in the applied load invalidates every dimension and requires the whole calculation to be performed again. In academic practice this repetition consumes the majority of the time spent on a design problem, while the engineering judgement involved, namely the choice of allowable stresses and the interpretation of the result, occupies very little of it. The drafting that follows the calculation is similarly mechanical.

Computer aided design software removes the labour of drawing but not the labour of calculation. A parametric modeller will faithfully regenerate a model when its driving parameters change, and will not object if the pin of a knuckle joint is too slender to carry the bending moment placed upon it. Determining the parameters remains the responsibility of the designer and remains manual.

## 1.2 Motivation

The motivation for this work is the gap between stating a requirement and obtaining a verified design. At present a designer must translate a requirement such as *a knuckle joint to carry 50 kN* into a sequence of formulas by hand, evaluate them, check the result against several failure modes, iterate wherever a check fails, and only then begin modelling and drafting.

A system that accepts the requirement as written and returns a verified design, complete with solid models and a drawing sheet, would remove the repetition without removing the engineering. The difficulty is that interpreting a written requirement is not a calculation, while the calculation itself must not be delegated to a mechanism that cannot be checked.

## 1.3 Objectives

The objectives of the present work are as follows.

1. To develop a system that accepts a design requirement stated in ordinary English and extracts from it the joint type, the applied load and the allowable stresses.

2. To size the joint using established design formulas and to verify every applicable failure mode against the allowable stress, iterating wherever a check is not satisfied.
3. To generate exact solid models of every component and of the assembly in a neutral exchange format that opens in commercial CAD software.
4. To generate a first angle projection drawing sheet carrying dimensions and a title block in conformity with Indian drawing practice.
5. To verify every proportion and every failure check used by the system against published design literature, and to record any disagreement with a source explicitly.
6. To make the system usable through a web browser without any installation.

### 1.4 Scope of the Work

The scope of the present work is limited to six standard joints under static loading. Fatigue loading, stress concentration factors, tolerances and fits, and the selection of manufacturing processes are outside the scope. The finite element facility included in the system performs a single linear static analysis on one component and is intended as an independent cross check on the formula rather than as a substitute for a complete analysis.

### 1.5 Organisation of the Report

Chapter 2 reviews the design literature consulted and the existing approaches to design automation. Chapter 3 identifies the gap and formulates the problem. Chapter 4 describes the joints selected as case studies and the data collected for them. Chapter 5 sets out the analysis and design methodology, including the division of responsibility between the language model and the deterministic calculation. Chapter 6 describes the implementation. Chapter 7 presents the results, the verification against published sources and the discrepancies found in those sources. Chapter 8 concludes and identifies the scope for further work.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Design of Cotter and Knuckle Joints

The primary reference used in this work is the course material on the Design of Machine Elements published by the Indian Institute of Technology Kharagpur through the National Programme on Technology Enhanced Learning. Module 4 of that course treats pin joints, keys and threaded fasteners. Lesson 2 gives the complete design procedure for the socket and spigot cotter joint and for the knuckle joint, including the empirical proportions expressed as multiples of the rod diameter and the full list of failure modes to be verified.

Two further texts were used, both out of copyright and obtained from the Internet Archive. Unwin, in *The Elements of Machine Design* of 1899, treats the gib and cotter joint on page 179 and gives the division of the slot breadth between the cotter and the gib, the thickness of both as a fraction of that breadth, and the permissible range of taper. Leutwiler, in *Elements of Machine Design* of 1917, gives in Article 98 a list of ten distinct ways in which a cotter joint may fail, together with the expression governing each of them.

The availability of two independent treatments of the cotter joint, published eighteen years apart, proved significant to this work. Where the two sources agree, the expression adopted here may be regarded as settled. Where they disagree, the disagreement is itself a finding, and is reported in Section 7.4.

## 2.2 Design of Shaft Couplings

Leutwiler, in Article 275, presents the analysis of the flange coupling as three groups of failure: failure of the key by shear or by crushing, failure of the coupling bolts by shear or by crushing, and shearing of the flange at the hub. The expressions given there for the bolt diameter are adopted in the present work.

The same article records that no uniform set of proportions for flange couplings had been proposed for adoption as a standard, and that the published proportion tables represent the practice of individual manufacturers. That observation directly limits what may be claimed about the proportions used in the present work, and the limitation is stated in Section 7.3 rather than passed over. The two key equations, shear across the whole key and bearing across half the key thickness, are taken from Module 4 Lesson 1 of the same NPTEL course.

## 2.3 Standards Consulted

Indian Standards were consulted for drawing practice and for the fasteners and keys used in the joints implemented. SP 46 governs first angle projection and the title

block. IS 2048 covers parallel keys and keyways, IS 1363 and IS 1364 cover hexagon head bolts, screws and nuts, and IS 2693 covers cast iron flange couplings.

### 2.4 Existing Approaches to Design Automation

Parametric modelling is standard in commercial CAD software. A model driven by a table of parameters regenerates when those parameters change, which removes the drafting labour associated with a change of size. It does not, however, determine the parameters. The designer must still compute them, and the software will regenerate an unsafe model as readily as a safe one.

Recent work applying large language models to computer aided design has largely taken the form of generating modelling code directly from a description. Such systems produce geometry quickly, but the dimensions they produce cannot be traced to a design calculation, and there is no mechanism by which a failure mode is verified. For an academic or a professional design task, geometry that cannot be justified is of limited value.

A third group of approaches embeds engineering knowledge in expert systems and design handbooks implemented as software. These are reliable within the range for which they were written, but the calculation is typically hidden from the user, so a result cannot be checked against a source without access to the implementation.

# CHAPTER 3

# GAP IDENTIFICATION AND PROBLEM FORMULATION

### 3.1 Gap Identified

From the review presented in Chapter 2, the gap lies neither in modelling nor in language interpretation, both of which are mature, but in the connection between them. What is absent is a system in which a written requirement is interpreted by a language model, the resulting parameters drive a deterministic design calculation taken from published literature, and the calculation itself remains open to inspection by the user.

Three specific deficiencies follow from this. First, parametric CAD regenerates geometry but verifies no failure mode. Second, systems that generate modelling code from a description produce dimensions that cannot be attributed to any formula. Third, where design calculation is automated, the working is generally hidden, so the result must be trusted rather than checked.

### 3.2 Problem Formulation

The problem addressed in this work is stated as follows. Given a design requirement expressed in ordinary English, produce a complete design of the specified machine joint such that every dimension is obtained from a published design formula or from a named failure check, every applicable failure mode is verified against its allowable stress, the geometry is generated in a neutral exchange format, a first angle drawing sheet is produced, and the entire working is reported to the user in a form that permits independent verification.

### 3.3 Conditions the Solution Must Satisfy

1. No dimension may be produced without an attributable basis, which is either a design formula with a cited source or an explicitly named failure check.
2. The language model may interpret the requirement but may not select any dimension and may not perform any check.
3. Any numerical value returned by the language model that does not appear in the original requirement must be discarded, so that an invented figure cannot enter the calculation.
4. Every applicable failure mode must be evaluated on every run and its arithmetic reported, not merely its outcome.
5. Where the empirical proportions used have no supporting source, the system must state that fact rather than imply a source it does not have.

# CHAPTER 4

# ABOUT THE CASE STUDY AND DATA COLLECTION

### 4.1 Selection of Case Studies

Six joints were selected. They were chosen because each is a standard joint with a published design procedure, because between them they exercise every common failure mode (tension, double shear, crushing, bending and torsion), and because they differ sufficiently in geometry to test whether the system generalises. Table 4.1 lists them.

**Table 4.1: Joints selected as case studies, with the duty adopted for each**

| Joint | Components | Dimensions | Failure checks | Duty adopted |
|---|---|---|---|---|
| Knuckle joint | 5 | 12 | 9 | Load P = 50,000 N |
| Socket and spigot cotter joint | 3 | 14 | 10 | Load P = 35,000 N |
| Sleeve and cotter joint | 5 | 12 | 8 | Load P = 50,000 N |
| Gib and cotter joint | 4 | 20 | 8 | Load P = 50,000 N |
| Hex bolt, nut and washer | 3 | 11 | 2 | Load P = 20,000 N |
| Protected type rigid flange coupling | 12 | 15 | 7 | Torque T = 500,000 N mm (500 N m) |

The flange coupling differs from the remaining five in that it is sized from a transmitted torque rather than from an axial load, and in that the number of bolts varies with the shaft diameter. It was included specifically to confirm that the system is not limited to axially loaded joints.

### 4.2 Data Collected

For each joint, three categories of data were collected from the sources listed in Chapter 2: the empirical proportions expressed as multiples of the governing dimension, the complete list of failure modes with the expression governing each, and the worked numerical example given in the source, which was later used to verify the implementation.

The allowable stresses and the preferred size series adopted are given in Table 4.2. The preferred size series is significant: the source material directs that a computed diameter be rounded up to the next standard size rather than used as computed, and the system follows that direction.

**Table 4.2: Allowable stresses adopted, and the sizes to which computed diameters are rounded up**

| Quantity | Value adopted | Source |
|---|---|---|
| Allowable tensile stress | 75 MPa | working value used in the source material |
| Allowable shear stress | 60 MPa | working value used in the source material |
| Allowable crushing stress | 150 MPa | working value used in the source material |
| Design factor | 1.25 | sized for 1.25 times the load, reported at the load |
| Preferred diameters | 6 to 22 in steps of 2 mm, 25 to 60 in steps of 5 mm, thereafter 10 mm | standard shaft sizes, NPTEL Module 4 Lesson 2 |

### 4.3 Basis for Verification

The worked examples given in the sources form the basis on which the implementation was verified. For the knuckle joint and the cotter joint, the source provides a complete numerical example with the load, the allowable stresses and every resulting dimension. Those examples were reproduced by the system and compared value by value. The outcome of that comparison, including the instances in which the system and the source disagree, is reported in Chapter 7.

# CHAPTER 5

# ANALYSIS AND DESIGN METHODOLOGY

### 5.1 Overall Approach

The system is organised as five stages executed in sequence. The written requirement is interpreted into parameters; the joint is sized; every failure mode is verified; the geometry is generated; and the drawing sheet is produced. Each stage passes a definite result to the next, and the intermediate result of every stage is reported to the user.

### 5.2 Role of the Language Model

The language model is used at the first stage only. It receives the written requirement and returns the joint type, the applied load in newtons, and any allowable stresses stated. It selects no dimension, applies no proportion and performs no check. Table 5.1 sets out the division of responsibility.

**Table 5.1: Division of responsibility between the language model and the deterministic calculation**

| Stage | Performed by | Verifiable against |
|---|---|---|
| Interpretation of the requirement | Language model | the requirement as written |
| Sizing of every dimension | Deterministic calculation | cited design formula |
| Verification of failure modes | Deterministic calculation | cited design formula |
| Generation of solid geometry | Modelling kernel | the computed dimensions |
| Generation of the drawing sheet | FreeCAD | the assembled geometry |

This division is deliberate and is the principal design decision of the project. Determining that the phrase *50 kN* denotes a load of 50,000 N, and that *socket and spigot* names the cotter joint, is a problem of language interpretation, at which such a model is effective. Determining the diameter of the pin is a problem of engineering with a single correct answer obtainable from a formula, and it is therefore performed by deterministic code whose every step is traceable to a published page.

A safeguard is applied to the output of the model. Any numerical value it returns that does not appear in the original requirement is discarded and the user is informed. This prevents a plausible but invented figure from entering the calculation. Where no language model is available, the system falls back to pattern matching rules and reports which interpreter was used.

### 5.3 Sizing Procedure

The governing dimension is computed first from the primary failure mode. For a knuckle joint this is the rod diameter obtained from direct tension, given by Equation (5.1).

$$D = \sqrt{( 4P / ( \pi \sigma_t ) )} \qquad (5.1)$$

The computed value is then rounded up to the next standard preferred size rather than used as computed, in accordance with the practice given in the source material.

Where a component has more than one independent failure mode, as the pin of a knuckle joint has in both double shear and bending, that component is sized separately from each applicable expression, given by Equations (5.2) and (5.3), and the larger requirement is adopted.

$$d = \sqrt{( 2P / ( \pi \tau ) )} \qquad (5.2)$$

$$d = [ 16P ( t_1/3 + t/4 ) / ( \pi \sigma_b ) ]^{1/3} \qquad (5.3)$$

The remaining dimensions follow from the empirical proportions given in the source, expressed as multiples of the governing dimension.

### 5.4 Design Factor

A design factor of 1.25 is applied. The joint is sized for 1.25 times the stated load, and the resulting stresses are then reported at the actual load. The consequence is that the governing check reports approximately 75 to 80 percent of the allowable stress rather than 100 percent, so that margin remains in hand. Sizing exactly to the allowable produces a design that satisfies every check on paper while retaining nothing for material variation or for overload.

### 5.5 Verification of Failure Modes

Every applicable failure mode is evaluated and compared with the corresponding allowable stress. Where a check is not satisfied, the dimension responsible is increased by one millimetre and the entire check set is evaluated again. The process repeats until all checks are satisfied. The dimension that was increased, and the check that caused the increase, are both reported to the user.

### 5.6 Generation of Geometry and Drawings

Each component is generated as an exact solid using a parametric modelling kernel and is written in the STEP neutral exchange format, which opens in Siemens NX, SolidWorks, FreeCAD, Onshape and Fusion. The assembly is written as a further STEP file with each component in its assembled position.

The drawing sheet is produced by driving FreeCAD without its graphical interface. The assembled geometry is projected to obtain the front view, the left side view and the plan, arranged in first angle projection in accordance with Indian

practice, together with an isometric view, dimension lines and a title block. The same views are also written in DXF format.

### 5.7 Finite Element Cross Check

An optional finite element analysis is provided as an independent check on the calculated stress. One component is meshed and solved as a linear static problem using the CalculiX solver, and the resulting peak stress is compared with the value obtained from the formula. Where the computed peak exceeds the calculated value by more than half, the analysis is repeated on a mesh of half the element size. If the peak then moves by more than five percent the result is reported as not converged, since a peak that continues to change with mesh refinement indicates a stress concentration at a sharp corner or at a constrained face rather than a stress to which a component may be designed.

# CHAPTER 6

# IMPLEMENTATION

### 6.1 System Architecture

The design rules for each joint are held as data rather than as code. Each joint is described by a file recording its empirical proportions, the source of each proportion, the default allowable stresses and the preferred size series. The calculation code reads these files. Two consequences follow. The source of any dimension can be reported to the user at the moment it is computed, and a correction to a proportion is a change to data rather than to code.

### 6.2 Software Tools

Table 6.1 lists the tools used. All of them other than the language model are open source.

**Table 6.1: Software tools used in the implementation**

| Function | Tool | Remark |
|---|---|---|
| Solid modelling | CadQuery 2.8 on the OCCT kernel | open source, scripted parametric modelling |
| Drawing sheet and DXF | FreeCAD, without its interface | projection, dimensioning and title block |
| Finite element analysis | CalculiX with gmsh | linear static, single component |
| Interpretation of the requirement | Claude language model | parameter extraction only |
| Application | Python standard library | no external web framework |
| Deployment | container on a public host | no installation required of the user |

### 6.3 Implementation of the Joints

Each joint is implemented as a module exposing the same interface: a sizing function, a list of failure checks, a geometry builder and a set of dimension specifications for the drawing sheet. Adding a joint therefore requires no change to the surrounding system, which is what allowed the gib and cotter joint to be added after the other five were complete.

The gib and cotter joint requires particular comment. Unwin gives the division of the breadth and the thickness of the gib and cotter but gives no proportions for the strap. Rather than adopt a ratio from an unrelated source or invent one, every dimension the source does not supply is sized against a failure check until that check is satisfied. The result is that no dimension of that joint is unsupported: each is either taken from the source or derived from a stated failure mode.

### 6.4 User Interface

The system is used through a web browser. The user enters the requirement, confirms or corrects the parameters extracted from it, and requests the build. The complete working of the calculation is printed as it proceeds. The results are presented as the three dimensional model, the drawing sheet, the list of dimensions, the list of checks, a view colouring components by utilisation, and the generated files.

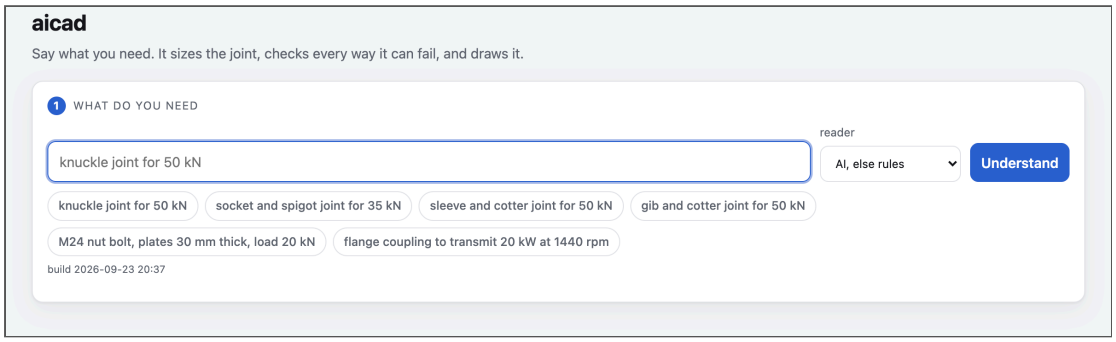



**Figure 6.1: Entry screen of the system, showing the requirement field and the example requirements provided**

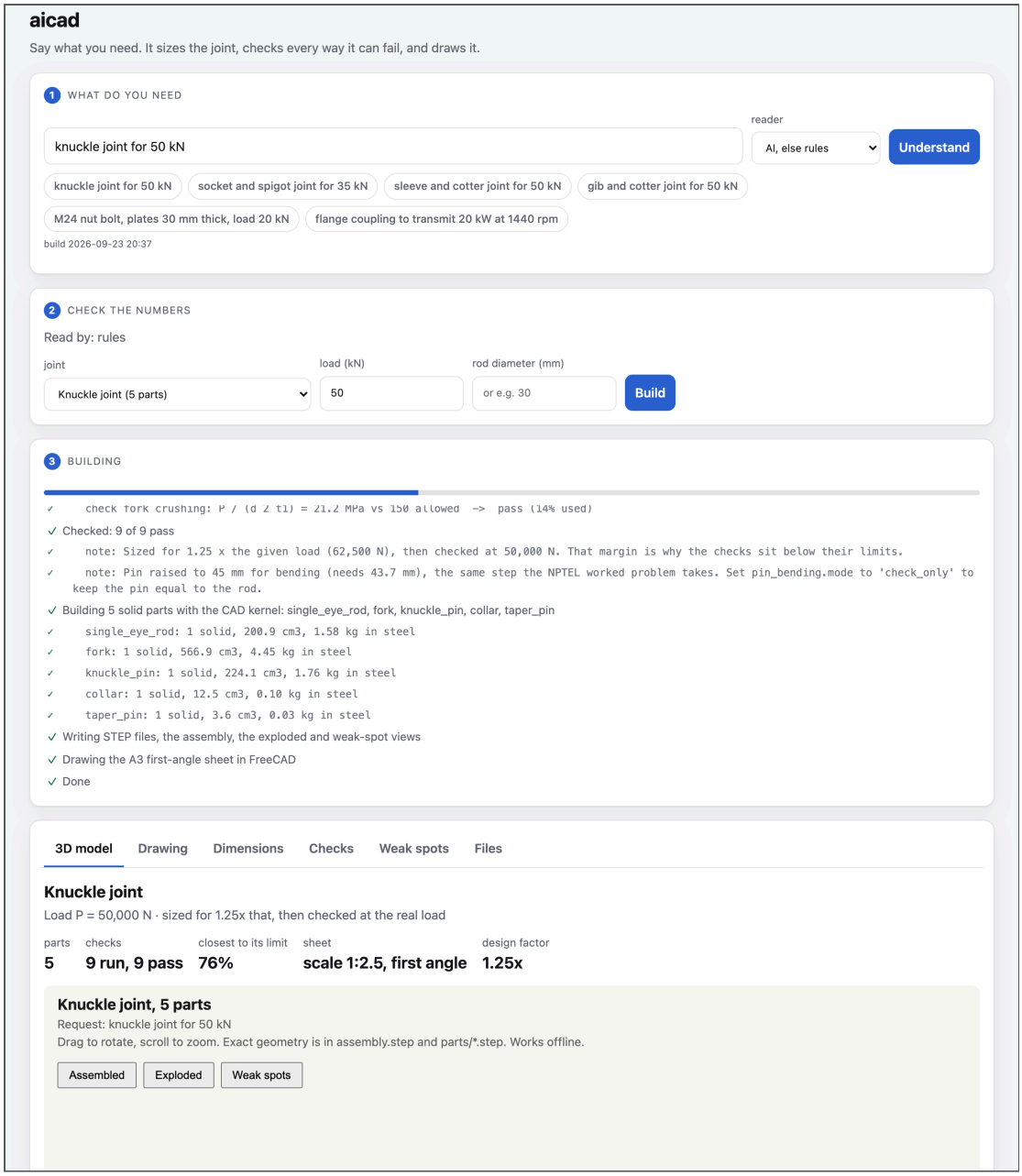



**Figure 6.2: Working printed during a single run, showing the values computed before rounding, the rule and source of each proportion, and every check with its arithmetic substituted**

## 6.5 Verification of the Implementation

The implementation is covered by 290 automated tests which are executed on every change. These cover the sizing arithmetic, the generated geometry, the interpretation of requirements, the drawing sheet and the finite element facility. Among them are tests which assert that the discrepancies recorded in Section 7.4 are genuine, so that the system cannot subsequently be altered to reproduce the printed answers without a test failing.

One class of test deserves mention because it detected an error that no dimensional check could have detected. The geometry tests compare the solids of every pair of components and require that no two occupy the same space. That test found a hooked gib modelled 1.25 mm inside the strap, an error invisible in the dimension list because every individual dimension was correct.

# CHAPTER 7

# RESULTS AND DISCUSSION

### 7.1 Design Results

All six joints size correctly, satisfy every applicable failure check, generate every component as a solid model and produce a drawing sheet. Each design completes in approximately ten seconds. Table 7.1 summarises the results.

**Table 7.1: Design results for the six joints at the duties given in Table 4.1**

| Joint | Components | Checks | Busiest check, percent of allowable | Outcome |
|---|---|---|---|---|
| Knuckle joint | 5 | 9 | 76 | all satisfied |
| Socket and spigot cotter joint | 3 | 10 | 67 | all satisfied |
| Sleeve and cotter joint | 5 | 8 | 77 | all satisfied |
| Gib and cotter joint | 4 | 8 | 79 | all satisfied |
| Hex bolt, nut and washer | 3 | 2 | 71 | all satisfied |
| Protected type rigid flange coupling | 12 | 7 | 79 | all satisfied |

The busiest check across the six joints lies between 67 and 79 percent of the allowable stress. That range is the intended consequence of the design factor described in Section 5.4 and confirms that margin is retained in every case.

### 7.2 Worked Example

The requirement *knuckle joint for 50 kN* is interpreted as the knuckle joint carrying an axial load of 50,000 N with the default allowable stresses. Applying the design factor, the joint is sized for 62,500 N. Table 7.2 gives the three governing calculations.

**Table 7.2: Governing calculations for a knuckle joint carrying 50 kN, before rounding**

| Expression | Computed, mm | Effect |
|---|---|---|
| Rod in direct tension | 32.57 | rod diameter adopted as 35 mm |
| Pin in double shear | 25.75 | not governing |
| Pin in bending | 43.72 | governs; pin diameter adopted as 45 mm |

The requirement arising from bending is nearly 1.7 times that arising from shear. A pin sized from shear alone would satisfy its own check and bend in service. This is the reason the larger of the two requirements is adopted, and the source

material takes the same step. Table 7.3 gives the complete set of dimensions with the rule and the source behind each.

**Table 7.3: Dimensions computed for the knuckle joint, with the rule behind each**

| Symbol | Meaning | Value, mm | Rule and source |
|---|---|---|---|
| D | rod diameter | 35 | rod in tension: D = sqrt(4P / (pi sigma_t)), rounded up to a preferred size |
| d | pin diameter | 45 | governed by pin bending |
| d2 | eye outer diameter | 70 | d2 = 2.0 x D, the rod diameter (outer diameter of eye) [NPTEL IIT KGP M4L2 fig 4 |
| d3 | pin head and collar diameter | 52.5 | d3 = 1.5 x D, the rod diameter (pin head and collar diameter) [NPTEL IIT KGP M4L |
| t | eye thickness | 43.75 | t = 1.25 x D, the rod diameter (thickness of single eye) [NPTEL IIT KGP M4L2 fig |
| t1 | fork plate thickness | 26.25 | t1 = 0.75 x D, the rod diameter (thickness of each fork plate) [NPTEL IIT KGP M4 |
| t2 | pin head thickness | 17.5 | t2 = 0.5 x D, the rod diameter (thickness of pin head) [NPTEL IIT KGP M4L2 fig 4 |
| collar_t | collar thickness | 22.5 | - |
| taper_d | taper pin diameter (big end) | 8.75 | - |
| clearance | gap between eye and fork base | 4.5 | - |

With all dimensions determined, the nine failure modes are evaluated at the actual load of 50,000 N. The results are given in Table 7.4. Every check is satisfied, and the busiest, shear of the eye, reaches 76 percent of its allowable stress.

**Table 7.4: Failure checks for the knuckle joint, evaluated at the actual load of 50 kN**

| Failure mode | Expression | Stress, MPa | Allowable, MPa | Utilisation, percent | Result |
|---|---|---|---|---|---|
| rod tension | `4P / (pi D^2)` | 52 | 75 | 69 | safe |
| pin double shear | `P / (2 pi/4 d^2)` | 15.7 | 60 | 26 | safe |
| pin bending | `32M / (pi d^3), M = P/2 (t1/3 + t/4)` | 55 | 75 | 73 | safe |
| eye tension | `P / ((d2 - d) t)` | 45.7 | 75 | 61 | safe |
| eye shear | `P / ((d2 - d) t)` | 45.7 | 60 | 76 | safe |
| eye crushing | `P / (d t)` | 25.4 | 150 | 17 | safe |
| fork tension | `P / ((d2 - d) 2 t1)` | 38.1 | 75 | 51 | safe |
| fork shear | `P / ((d2 - d) 2 t1)` | 38.1 | 60 | 64 | safe |
| fork crushing | `P / (d 2 t1)` | 21.2 | 150 | 14 | safe |

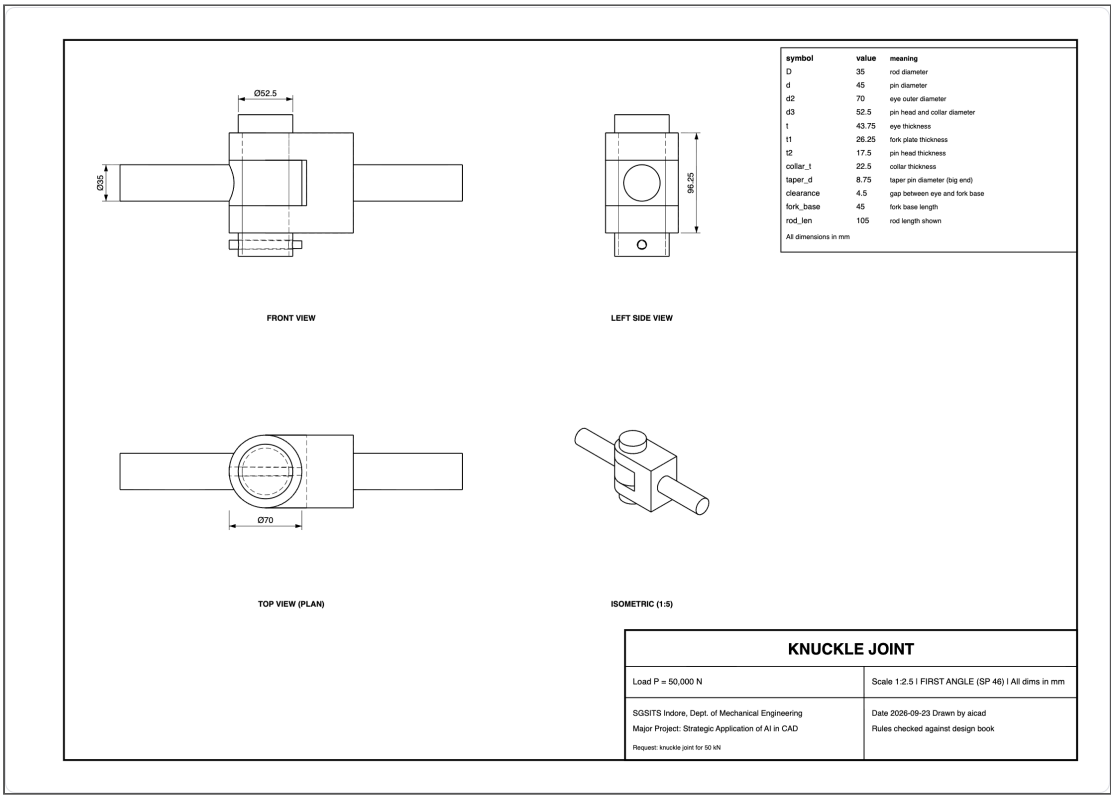



**Figure 7.1: First angle A3 drawing sheet generated for the knuckle joint of Section 7.2, at a scale of 1:2.5**

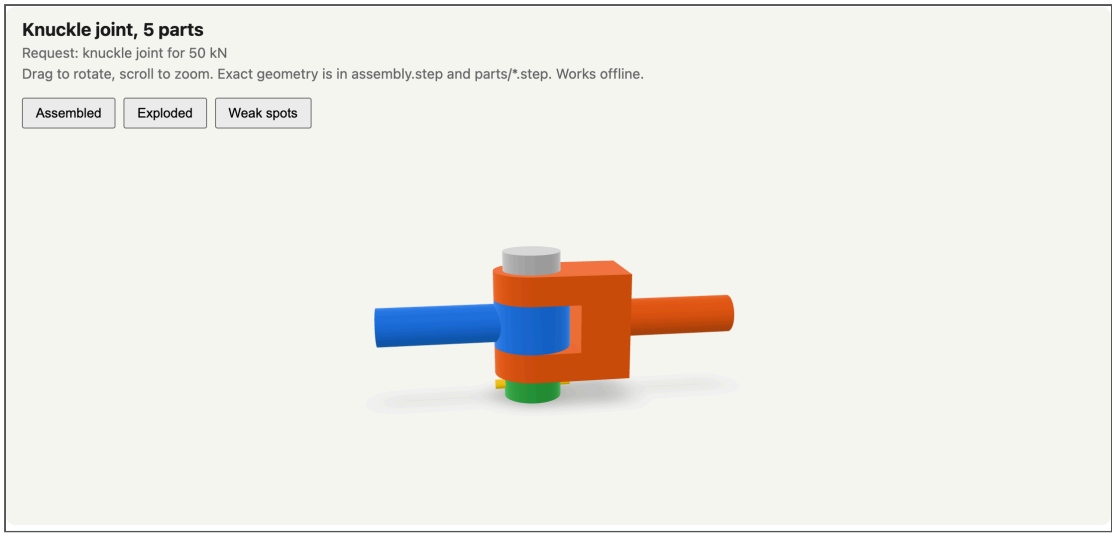



**Figure 7.2: Assembled model of the same joint, also available exploded and coloured by utilisation**

### 7.3 Verification Against Published Literature

Every proportion and every failure check was compared against the sources reviewed in Chapter 2. The outcome is given in Table 7.5.

**Table 7.5: Verification of each joint against published design literature**

| Joint | Source consulted | Outcome |
|---|---|---|
| Knuckle joint | NPTEL Module 4 Lesson 2 | proportions and all nine checks verified |
| Socket and spigot cotter joint | NPTEL Module 4 Lesson 2; Leutwiler Article 98 | all ten checks verified against two independent sources |
| Gib and cotter joint | Unwin, page 179 | breadth division, thickness and taper verified |
| Protected type flange coupling | NPTEL Module 4 Lesson 1; Leutwiler Article 275 | all seven checks verified twice; proportions not verifiable |
| Sleeve and cotter joint | four sources searched, none covers it | checks verified by correspondence; proportions unverified |
| Hex bolted joint | NPTEL Module 4 Lesson 4 | tightening relation verified; size table unverified |

Two entries require explanation. For the flange coupling, all seven failure checks are verified against two independent sources, but the empirical proportions are marked unverified. This is not an omission. As recorded in Section 2.2, Leutwiler states that no standard set of proportions was ever agreed and that published tables reflect the practice of individual manufacturers. The proportions used are therefore a convention, and the system reports them as such on every run. For the sleeve and cotter joint, four sources were searched and none treats that joint by name; its checks

are verified by direct correspondence with the socket and spigot cotter joint, but its proportions remain unsupported and are reported as such.

### 7.4 Discrepancies Found in the Sources

The verification exercise identified three errors in the published sources. They are recorded here because the behaviour of the system deliberately differs from the printed material in consequence.

#### 7.4.1 A Worked Example That Fails Its Own Check

The knuckle joint worked problem raises the pin diameter from 40 mm to 55 mm to satisfy bending, but subsequently evaluates the eye using the original 40 mm pin. Evaluated with the pin actually adopted, its own expression for shear of the eye gives 80 MPa against an allowable of 65 MPa, an exceedance of 23 percent. The present system instead enlarges the outer diameter of the eye and reports that it has done so.

#### 7.4.2 An Omitted Failure Mode

The cotter joint procedure in the same source verifies crushing of the socket but does not verify crushing between the cotter and the spigot. Evaluated at the final answer given in that source, the omitted check gives 167 MPa against an allowable of 110 MPa. Leutwiler, in Article 98, lists this mode as one of ten, and its expression is identical in form. The present system retains the check and sizes the cotter thicker in consequence. That two independent texts, published eighteen years apart, contain a check which the lesson omits is what establishes this as an omission rather than a matter of preference.

#### 7.4.3 A Constant Printed Incorrectly

The initial tightening load for a bolted joint is printed as $284d$ kN. The worked example on the same page obtains 56.8 kN for an M20 bolt, which is consistent only with a constant of $2840d$ newtons. The printed relation is therefore in error by a factor of ten, and the source refutes itself. The system uses $2840d$ N and reports the tightening load on every bolted design, since at M20 it amounts to 56,800 N before any service load is applied.

### 7.5 Finite Element Comparison

The finite element facility was first validated against a case with a known answer. Table 7.6 gives the comparison.

**Table 7.6: Comparison of the finite element result with the calculated stress**

| Case | Calculated, MPa | Computed, MPa | Difference, percent |
|---|---|---|---|
| Bar of 20 mm diameter, 10 kN in tension | 31.83 | 32.22 | +1.2 |
| Knuckle joint eye rod, 50 kN | 52.0 | 57.9 | +11.5 |

The bar validates the method. The eye rod reads higher because the calculation treats the rod as a plain bar in tension and takes no account of the stress raised at the pin bore, whereas the finite element model does. Both values remain below the allowable stress of 75 MPa.

The convergence rule described in Section 5.7 proved necessary in practice. For one component the computed peak varied between 157 and 191 MPa across a sixteen fold increase in the number of nodes without settling, which indicates a singularity at the constrained face rather than a physical stress. The system reports such a result as not converged and directs the user to the calculated value.

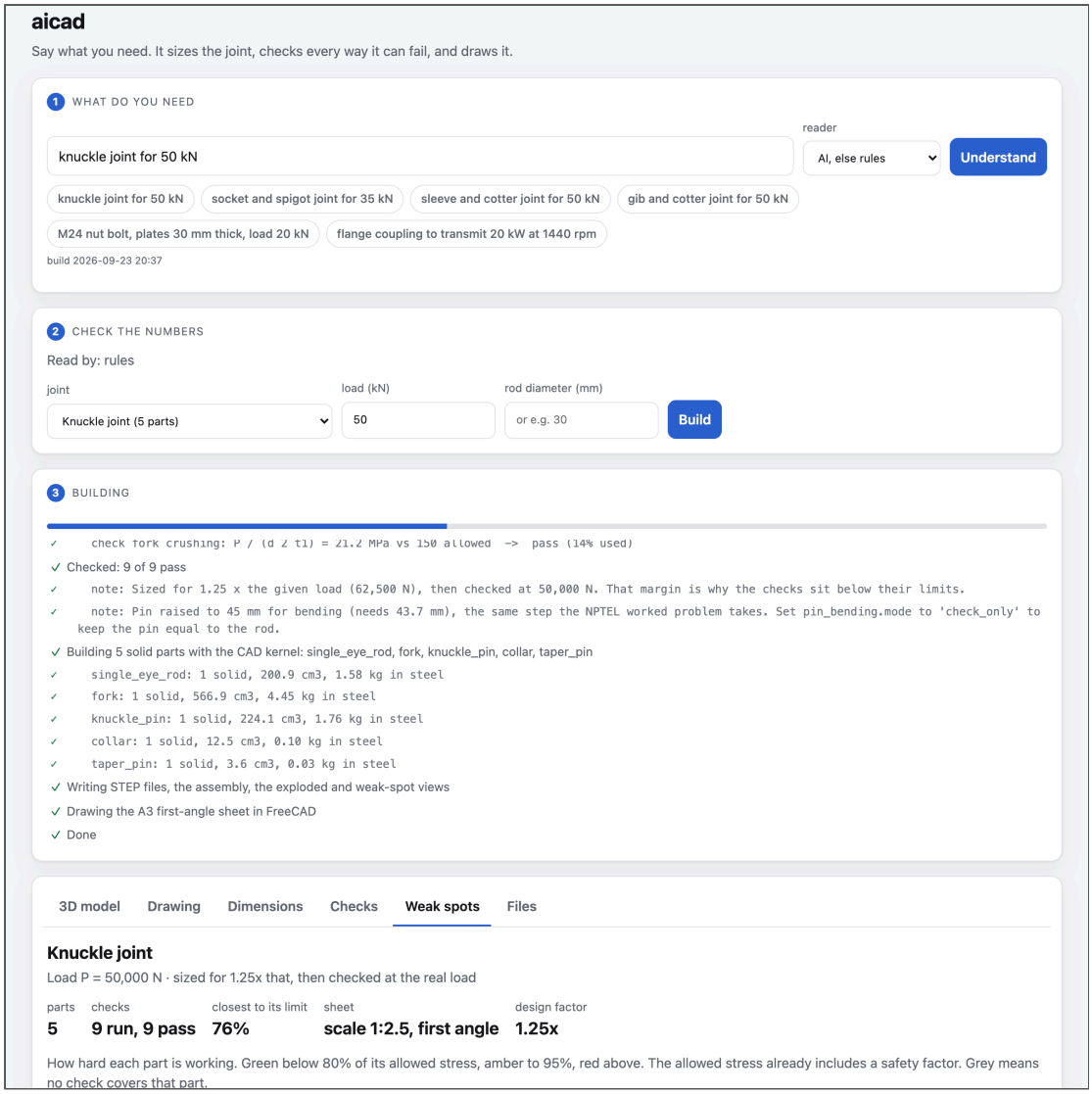



**Figure 7.3: Components coloured by utilisation, so that the member nearest its limit is apparent without reading the table**

## 7.6 Discussion

The results support the approach adopted. The division of responsibility set out in Section 5.2 allowed a natural language interface to be provided without any part of the design becoming unverifiable, and the requirement that every check be evaluated and its arithmetic printed is what made the three discrepancies of Section 7.4 visible. None of them was found by inspection of the sources; all three surfaced because a number computed by the system disagreed with a number printed in a book, and the disagreement could be traced.

# CHAPTER 8

# CONCLUSIONS AND SCOPE OF FUTURE WORK

## 8.1 Conclusions

A system has been developed which accepts a machine joint requirement written in ordinary English and returns a complete design: every dimension computed from published design formulas, every applicable failure mode verified against the allowable stress, every component generated as an exact solid model, and a first angle drawing sheet produced with dimensions and a title block. Six joints are implemented, applying 44 distinct failure checks in total, and the implementation is covered by 290 automated tests.

The objective of keeping the design traceable has been met. Every dimension the system produces can be attributed either to a formula with a cited source or to an explicitly named failure check. The working of each run is printed in full, which allows the result to be verified rather than accepted.

The verification exercise against published literature produced an unanticipated result. Three errors were identified in the sources themselves, one of which is a worked example whose own answer exceeds its own allowable stress by 23 percent. This was found not by inspection but because the system evaluates every check on every design and reports the arithmetic, which makes an omitted or inconsistent check visible immediately.

The division of responsibility adopted here, in which the language model interprets the requirement and deterministic code performs the engineering, is the element of the work that is most generally applicable beyond the six joints implemented.

## 8.2 Scope for Future Work

1. **Additional joints.** Riveted and welded joints, treated in Module 10 of the source material, follow the same pattern and could be added without change to the architecture.
2. **Fatigue loading.** All sizing at present is static. Introducing an endurance limit and a stress concentration factor would extend the system to cyclically loaded joints.
3. **Tolerances and fits.** The drawing sheet carries nominal dimensions only. Adding fits for the pin and its bore would make the output directly usable for manufacture.
4. **Verification of the remaining proportions.** Two of the six joints have proportions without a supporting source. A design data handbook would settle both.

5. **Direct integration with commercial CAD.** Output is presently exchanged as STEP. A journal script for Siemens NX would allow the assembly to be constructed within that software rather than imported into it.
6. **Manufacturing outputs.** Material selection and cost estimation are partially implemented and could be extended to produce a bill of materials.

# APPENDIX A

# SCREENSHOTS OF THE WORKING SYSTEM

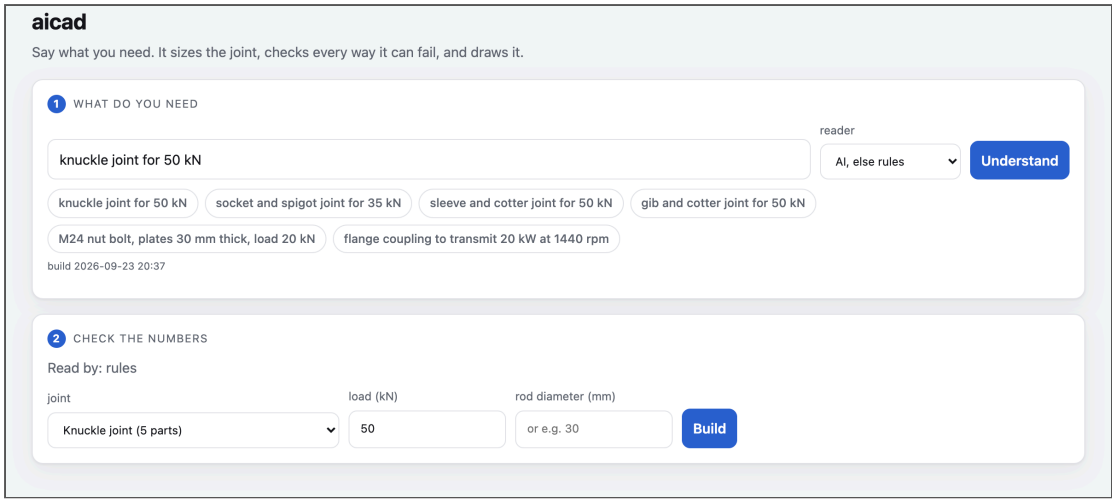



**Figure A.1: Parameters extracted from the requirement, presented for confirmation or correction before the design proceeds**

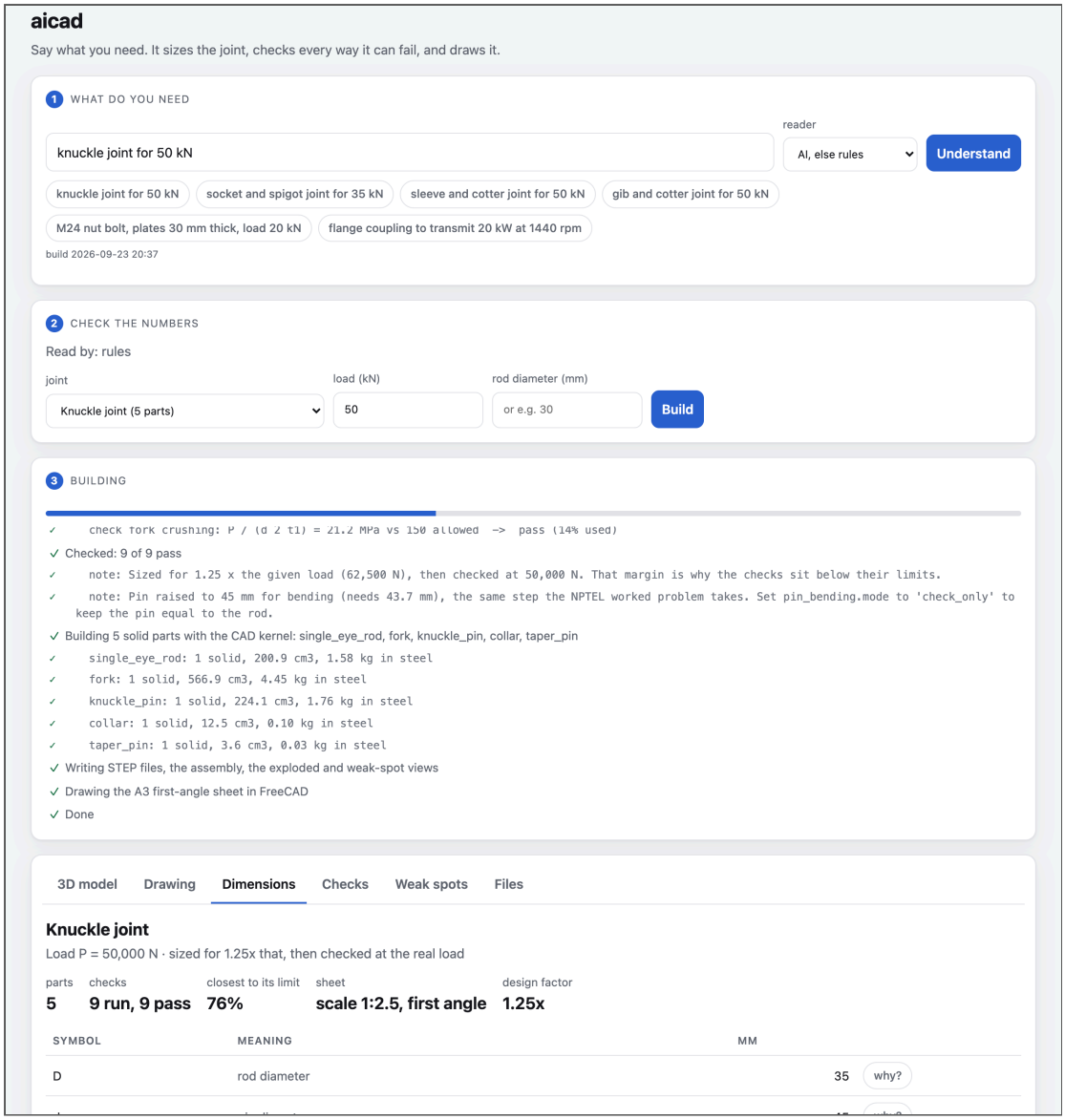



**Figure A.2: Every computed dimension with the rule that produced it and its source**

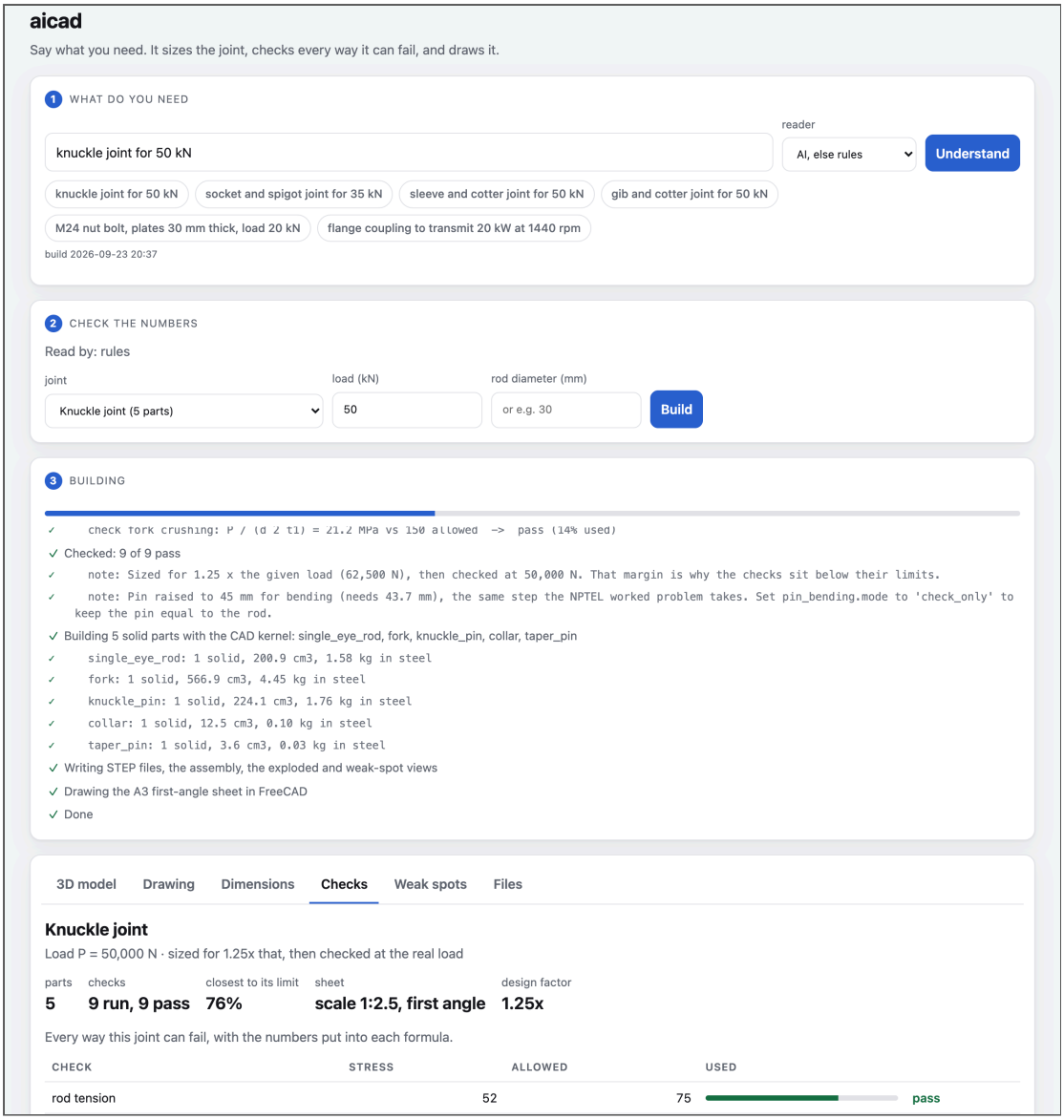



**Figure A.3: Every failure check with its substituted arithmetic, its allowable stress and its utilisation**

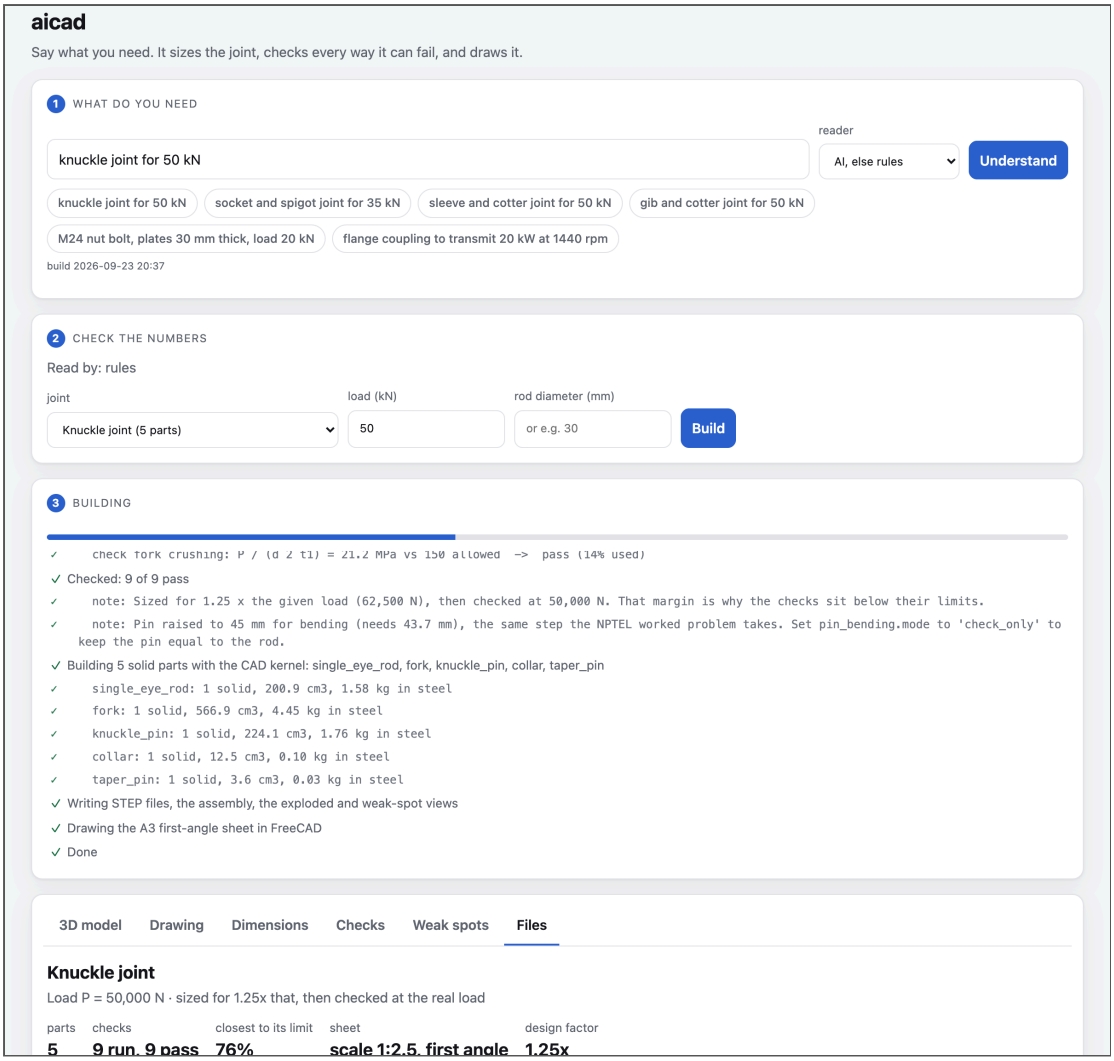



**Figure A.4: Files produced by a single run, available for download**

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